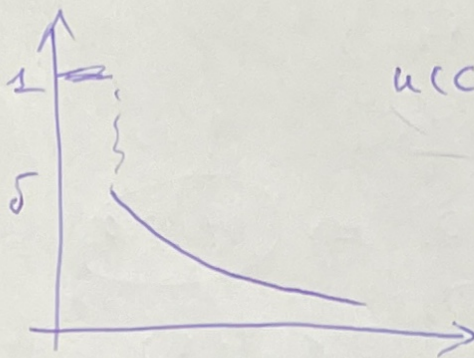
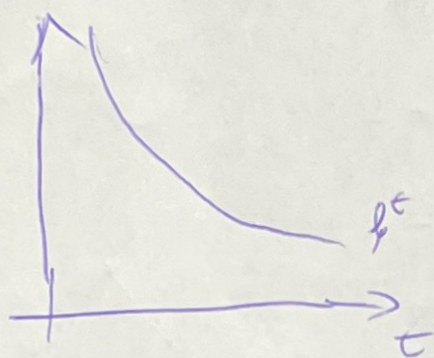




$$u(c_0) + \delta \sum_{t=1}^T \beta^t u(c_t)$$

$$\mathcal{L} = \sum_{t=0}^T \left[\beta^t u(c_t) + \lambda_t (R B_t + y_t - c_t - B_{t+1}) \right] + \mu \cdot B_{T+1}$$

$$\min_{\{\lambda_t\}} \max_{\{c_t, B_t\}} \mathcal{L}(c_t, B_t, \lambda_t, \mu)$$

$$\text{FOCs: wrt } c_t : \beta^t u'(c_t) - \lambda_t = 0$$

$$\text{wrt } B_{t+1} : -\lambda_t + R \lambda_{t+1} = 0 \quad t < T$$

$$\text{wrt } B_{T+1} : \mu - \lambda_T = 0 \quad t = T$$

CS:

$$\mu \cdot B_{T+1} = 0 \Rightarrow \lambda_T B_{T+1} = 0$$

$$\lambda_t = \beta^t u'(c_t) \Rightarrow \boxed{\beta^t u'(c_t) = R \cdot \beta^{t+1} u'(c_{t+1})} \quad (\text{EE})$$

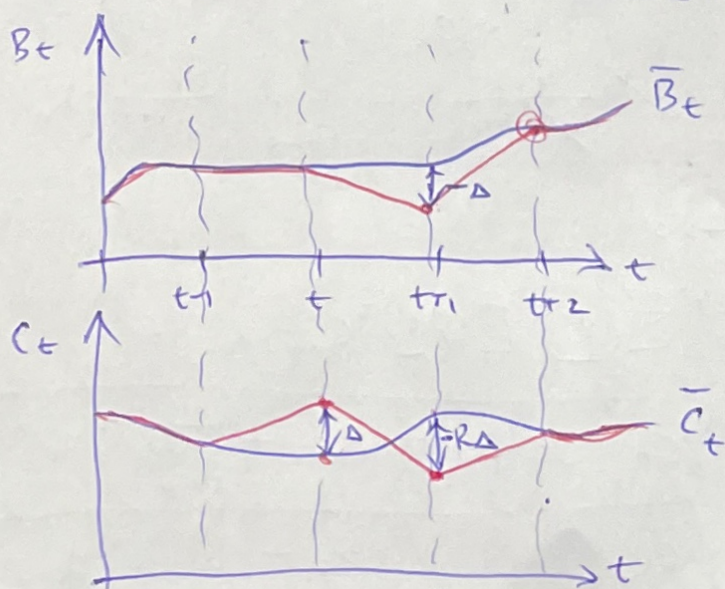
$$\boxed{\beta^T u'(c_T) B_{T+1} = 0} \quad (\text{TVC})$$

transversality condition

$$u'(c_t) = \beta R \cdot u'(c_{t+1})$$

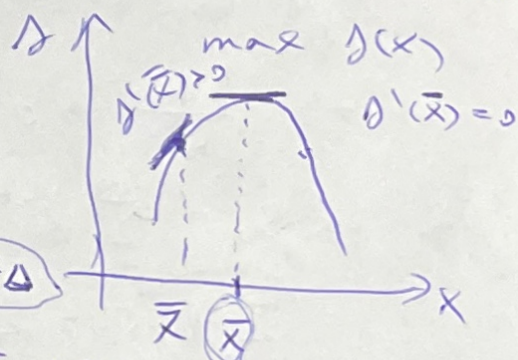
$$- \text{if } R=1 \Rightarrow u'(c_t) = u'(c_{t+1}) \Rightarrow c_t = c_{t+1}$$

$$- \text{if } R < 1 \Rightarrow u'(c_t) < u'(c_{t+1}) \Rightarrow c_t > c_{t+1} \Rightarrow \text{consumption smoothing}$$



$$\begin{aligned} d c_t &= \Delta \\ d B_{t+1} &= -\Delta \\ d B_{t+2} &= 0 \end{aligned}$$

$$\Rightarrow d c_{t+1} = -R \cdot \Delta$$



$$\Delta U = \sum_{t=0}^T \beta^t \Delta u(c_t) = \beta^t \Delta u(c_t) + \beta^{t+1} \Delta u(c_{t+1}) = \beta^t u'(c_t) \Delta + \beta^{t+1} (-R \Delta) u'(c_{t+1})$$

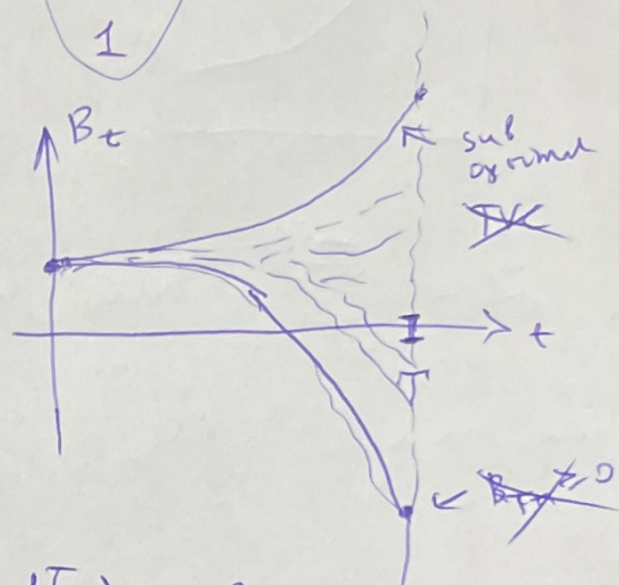
$$\begin{cases} u'(C_t) = \beta R u'(C_{t+1}) \\ B_{t+1} = R B_t + y_t - C_t \\ \beta^T u'(C_T) B_{T+1} = 0 \Rightarrow B_{T+1} = 0 \end{cases}$$

$$\begin{cases} \{C_t\} \rightarrow (T+1) \\ \{B_t\} \rightarrow (T+1) \end{cases} \quad 2 \times (T+1)$$

$$(B_0, C_0) \xrightarrow[BC]{EE} (B_1, C_1) \rightarrow \dots \rightarrow (B_T, C_T)$$

$$\xrightarrow[BC]{\quad} B_{T+1}$$

"shooting algorithm"



$$\begin{aligned} \beta^T u'(C_T) B_{T+1} &\rightarrow TVC \\ B_{T+1} &\rightarrow -\infty \rightarrow NPSC \end{aligned}$$

$$u(C) = \log C$$

$$(EE) \Rightarrow \frac{1}{C_t} = \beta R \frac{1}{C_{t+1}} \Rightarrow C_{t+1} = \beta R \cdot C_t \Rightarrow C_t = (\beta R)^t C_0$$

$$(BC) \quad B_{t+1} = R B_t + y_t - C_t \quad \xRightarrow{t=0} B_0 = \frac{C_0 - y_0}{R} + \frac{B_1}{R}$$

$$\xRightarrow{t=1} B_1 = \frac{C_1 - y_1}{R} + \frac{B_2}{R} \quad B_2 = \dots$$

$$\Rightarrow B_0 = \frac{C_0 - y_0}{R} + \frac{C_1 - y_1}{R^2} + \frac{B_2}{R^2} = \dots$$

$$= \sum_{t=0}^{\infty} R^{-t-1} (C_t - y_t) + \lim_{t \rightarrow \infty} \frac{B_t}{R^t}$$

$$(TVC) \quad \lim_{t \rightarrow \infty} \beta^t \frac{B_{t+1}}{C_t} \stackrel{(EE)}{=} \lim_{t \rightarrow \infty} \beta^t \frac{B_{t+1}}{\beta^t R^t C_0} = \lim_{t \rightarrow \infty} \frac{B_{t+1}}{R^{t+1}} \cdot R$$

$$\Rightarrow \frac{B_{t+1}}{R^{t+1}} \rightarrow 0 \Rightarrow \frac{B_t}{R^t} \rightarrow 0$$

$$\Rightarrow R B_0 = \sum_{t=0}^{\infty} R^{-t} (C_t - y_t) \stackrel{(EE)}{=} \sum_{t=0}^{\infty} R^{-t} [(\beta R)^t C_0 - y_t] =$$

$$= \underbrace{\sum_{t=0}^{\infty} \beta^t C_0}_{\frac{1}{1-\beta} \cdot C_0} - \sum_{t=0}^{\infty} R^{-t} y_t \Rightarrow \begin{cases} C_0 = (1-\beta) [R B_0 + \sum_{t=0}^{\infty} R^{-t} y_t] \\ C_t = (1-\beta) [R B_t + \sum_{j=0}^{\infty} R^{-j} y_{t+j}] \end{cases}$$

(3)

$$\underbrace{B_{t+\Delta}}_{\substack{\uparrow \\ B_{t+\Delta}}} = R \underbrace{B_t}_{\substack{\uparrow \\ B_t}} + \underbrace{y_t}_{\substack{\uparrow \\ y_t \Delta}} - \underbrace{C_t}_{\substack{\uparrow \\ C_t \Delta}}$$

$1+r \cdot \Delta$

$$\Rightarrow B_{t+\Delta} = (1+r\Delta) B_t + y_t \Delta - C_t \Delta$$

$$B_{t+\Delta} - B_t = (2B_t + y_t - C_t) \cdot \Delta$$

$$\frac{B_{t+\Delta} - B_t}{\Delta} = 2B_t + y_t - C_t$$

$$\frac{dB}{dt}$$

$$\Delta \rightarrow 0$$

$$\dot{B}_t = 2B_t + y_t - C_t$$